



جامعة نيويورك أبوظبي
NYU ABU DHABI

Advanced Circuits

ENGR-UH 2311 | Spring I - 2025

Experiment: **Lab 3**

Date: **04 / 02 / 2025**

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Lab 3: Transient Simulation of circuits involving Resistors, Capacitors, and Inductors in Cadence

Objectives:

This lab focuses on the transient response of RLC circuits using Cadence simulation software. The primary objectives include understanding the Cadence environment for circuit simulation, creating and figuring circuit schematics, defining and simulating transient responses for various circuit configurations, analysing simulation results through voltage, current, and power calculations and interpreting and documenting trends in transient response plots.

Diagram

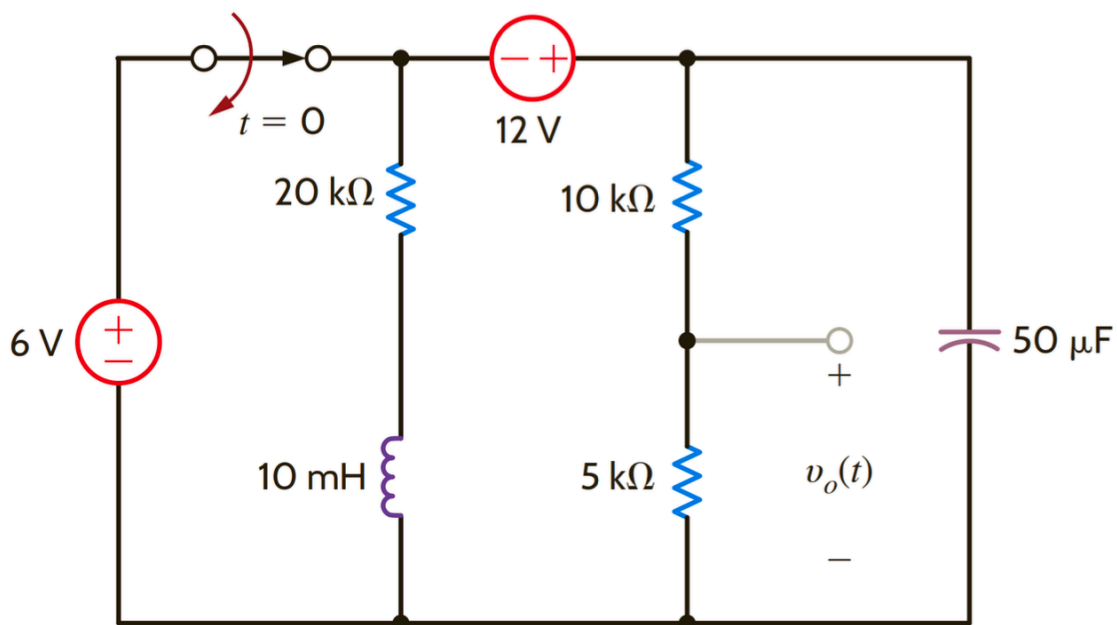


Figure 2: Circuit 2 Schematic having RLC and a switch

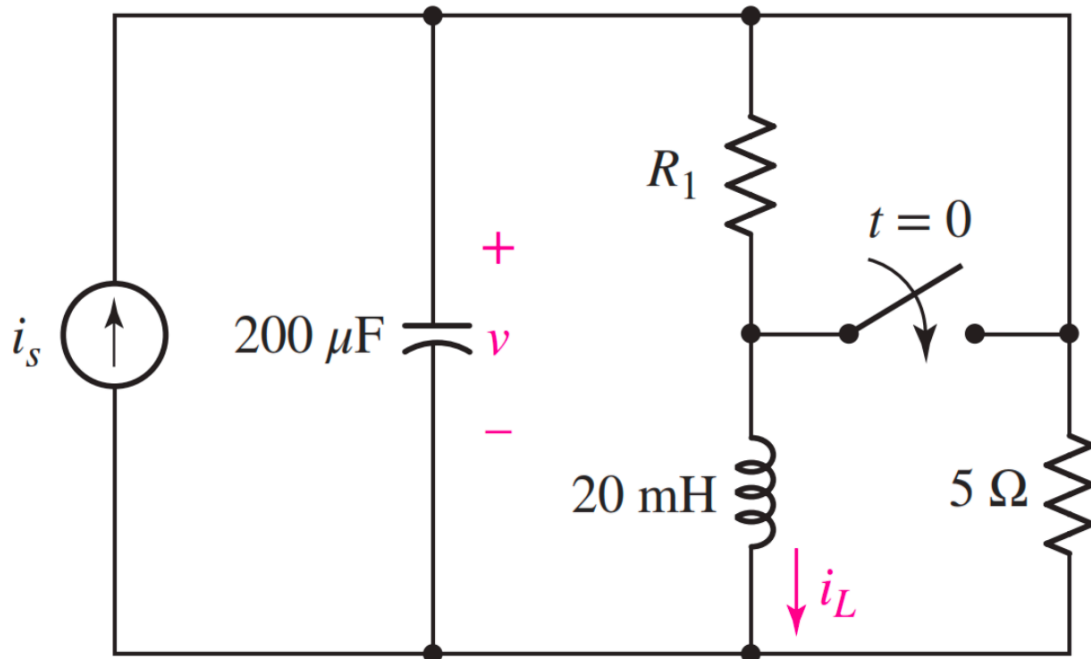
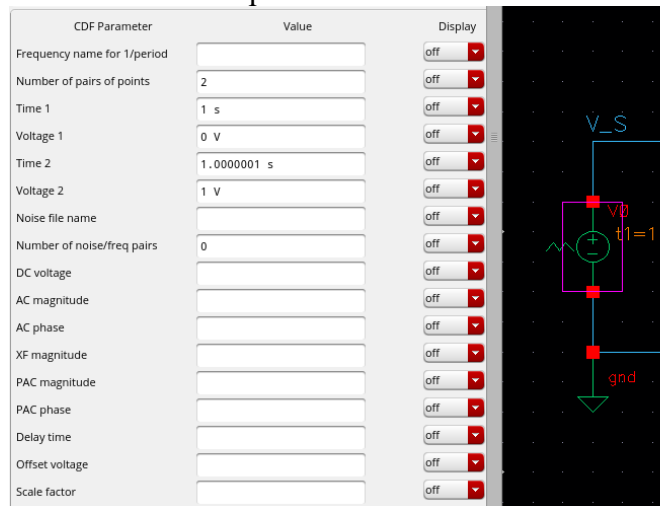


Figure 2: Circuit 3 Schematic involving RLC and a switch

Procedure

Circuit 1: RLC Series Circuit Simulation

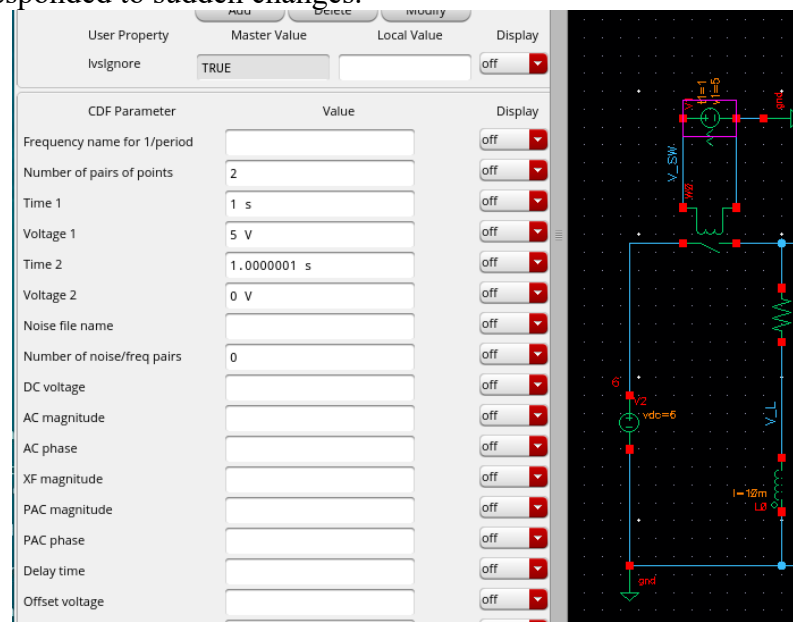
1. An RLC series circuit was built in Cadence using a resistor ($R = 5\Omega$), an inductor ($L = 1\text{mH}$), and a capacitor ($C = 1\mu\text{F}$) connected in series.
2. A unit step voltage source (0V to 1V) was applied to the circuit with zero initial conditions to observe the transient response.



3. The simulation environment was configured, and transient analysis was performed to capture the voltage and current waveforms over time.
4. The obtained simulation results were analysed to observe the oscillatory behaviour and damping effects in the circuit.
5. The resistor value was then modified to 0Ω , and the simulation was re-run to analyse the system's behaviour in the absence of resistance. The resulting waveforms were compared with the previous case to examine the effect of resistance on damping.

Circuit 2: Transient Response Analysis

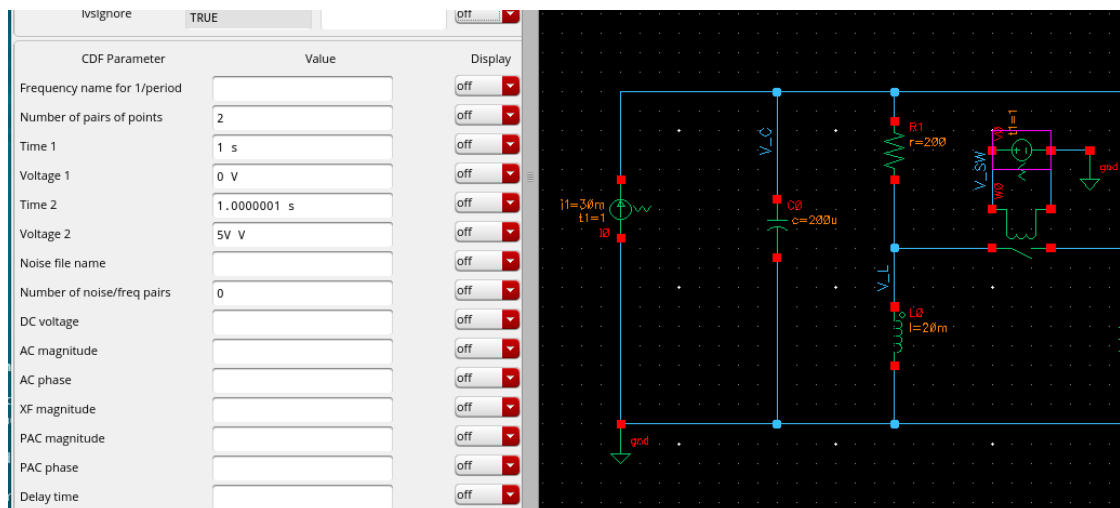
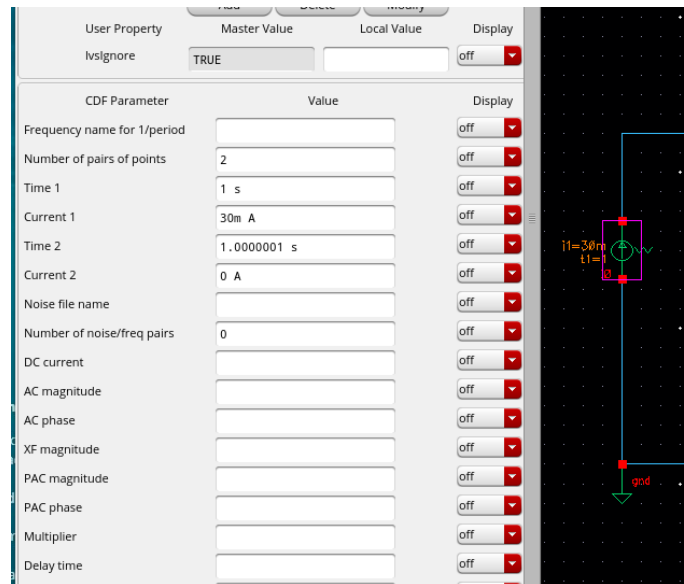
1. An RLC circuit with a switch was constructed in Cadence to study transient behaviour over a simulation period of 7 seconds.
2. The switch was initially open and then closed at time $t=1$ s to observe how circuit parameters responded to sudden changes.



3. The simulation was set up to measure voltages and currents at different time intervals for a comprehensive understanding of transient behaviour.
4. The collected data was analysed to determine settling time, peak values, and any oscillations or exponential decay trends.

Circuit 3: Analysis of $v_C(t)$ and $i_L(t)$

1. An RLC circuit with a defined current source was implemented in Cadence.



- The circuit was designed with a resistor, an inductor, and a capacitor.
- The transient simulation was run to observe the behaviour of capacitor voltage and inductor current over time.
- The settling time of the capacitor voltage and inductor current was analysed to determine if they followed the same response characteristics or exhibited different rates of stabilization.
- The simulation results were documented, and trends in the response curves were examined to understand the circuit's transient characteristics better.

Results & Observations

Circuit 1:

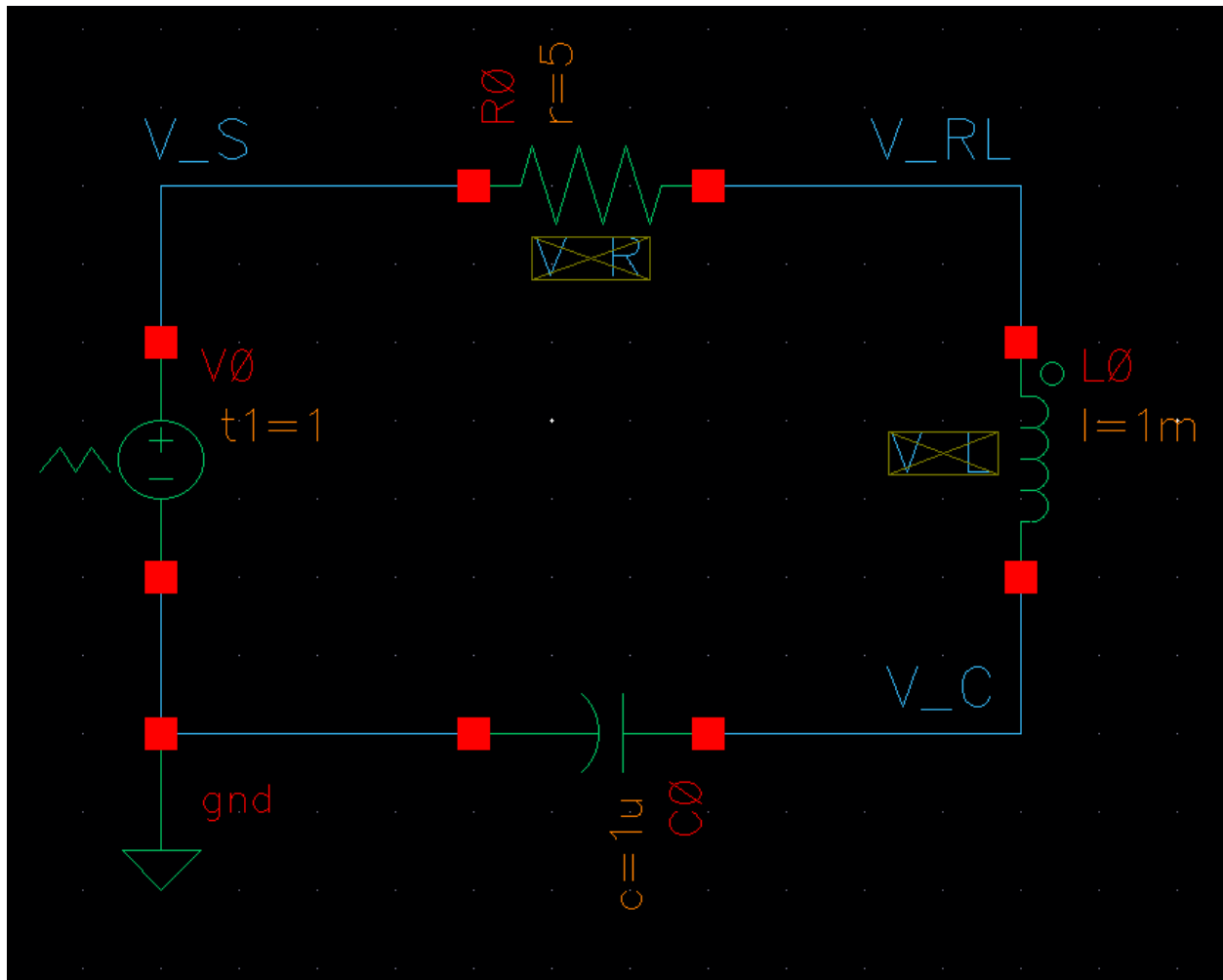


Figure 3: Schematic of Series RLC Circuit with 5 Ohm Resistor

With $R = 5\Omega$, the initial transient response showed oscillations with damping

Transient Response

Thu Feb 13 23:57:30
2025

1

Namet

- V_S
- V_{RL}
- V_C
- V_O/MINUS
- $/R_O/\text{PLUS}$
- $/L_O/\text{PLUS}$
- $/C_O/\text{PLUS}$

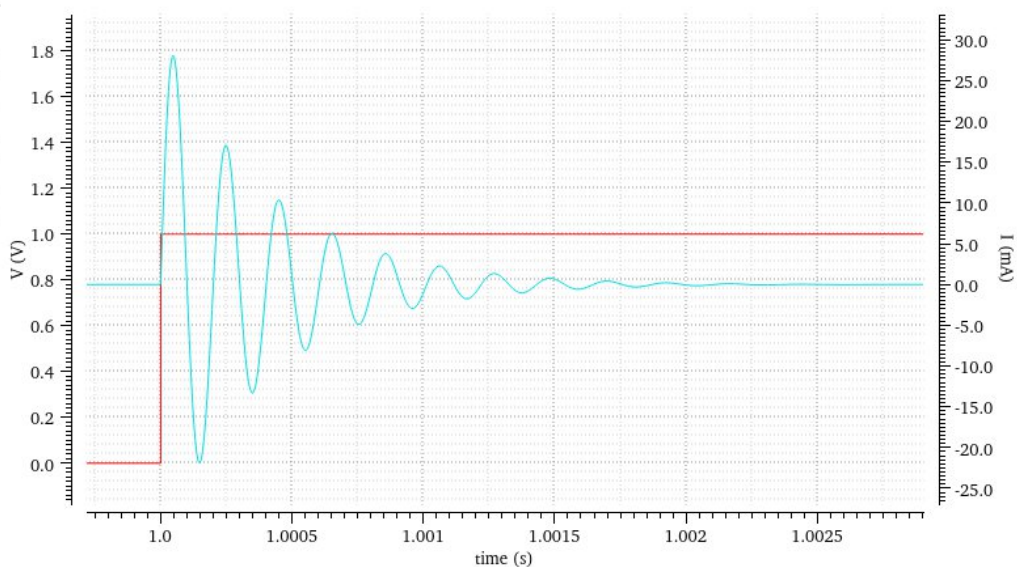


Figure 4. Graph for Transient Response of the Source Current and Voltage

Transient Response

Thu Feb 13 23:57:30
2025

1

Namet

- V_S
- V_{RL}
- V_C
- V_O/MINUS
- $/R_O/\text{PLUS}$
- $/L_O/\text{PLUS}$
- $/C_O/\text{PLUS}$

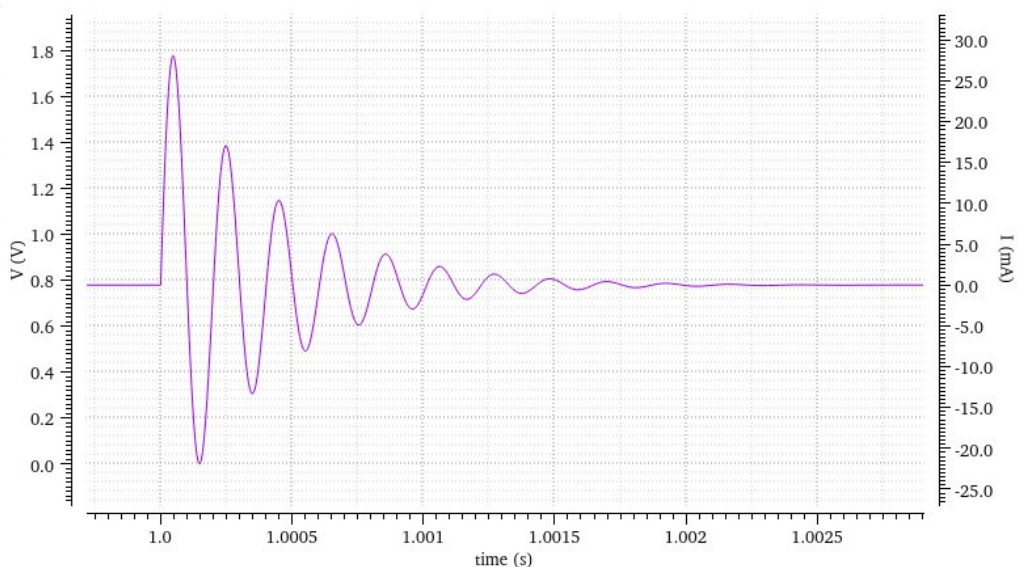


Figure 5. Graph for Transient Response of the Resistor

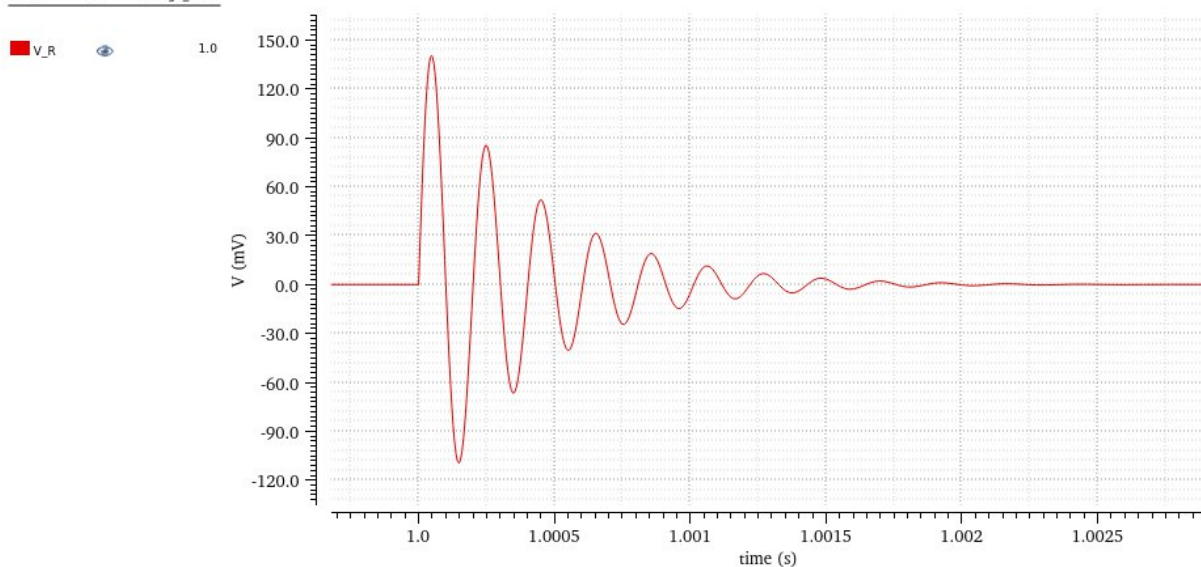


Figure 6. Graph for Transient Response for the Resistor Voltage

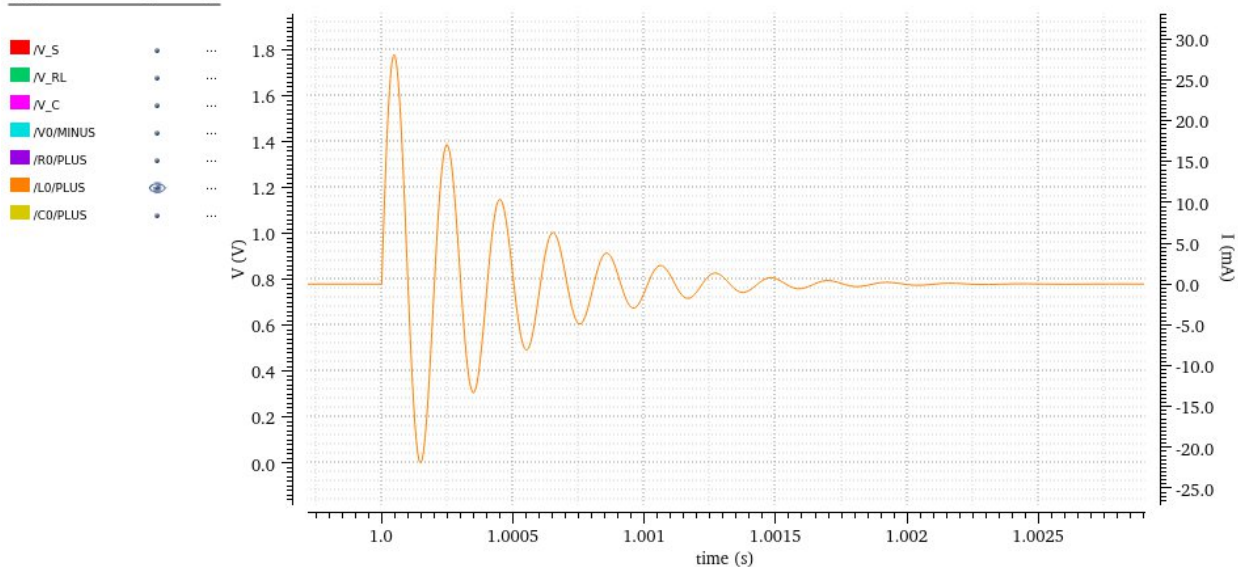


Figure 7. Graph for the Transient Response of the Inductor Current

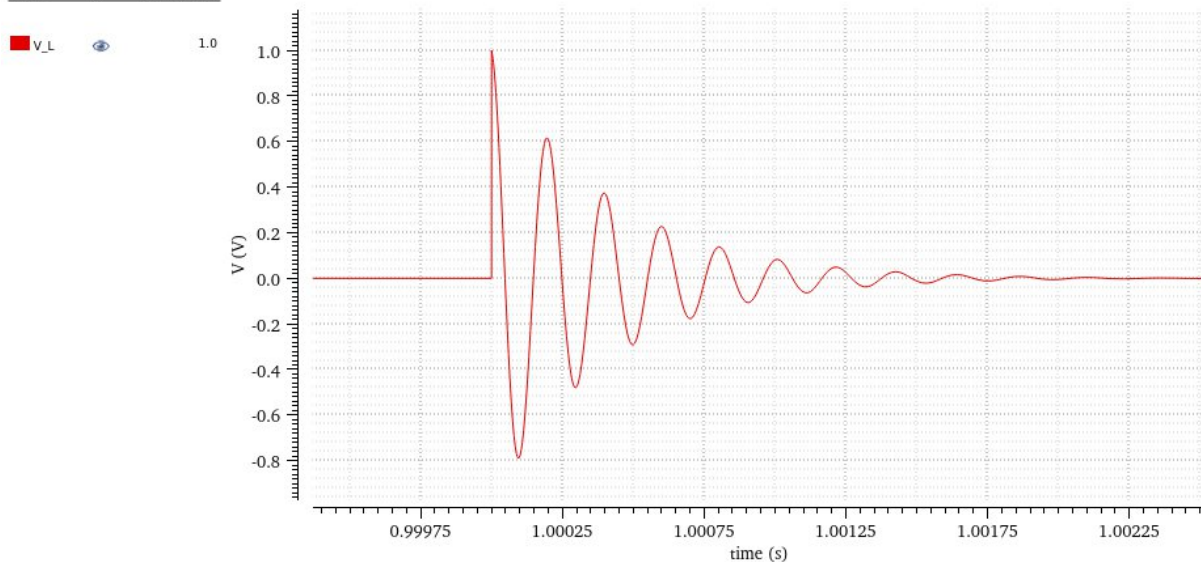


Figure 8. Graph for Transient response for the Inductor Voltage.

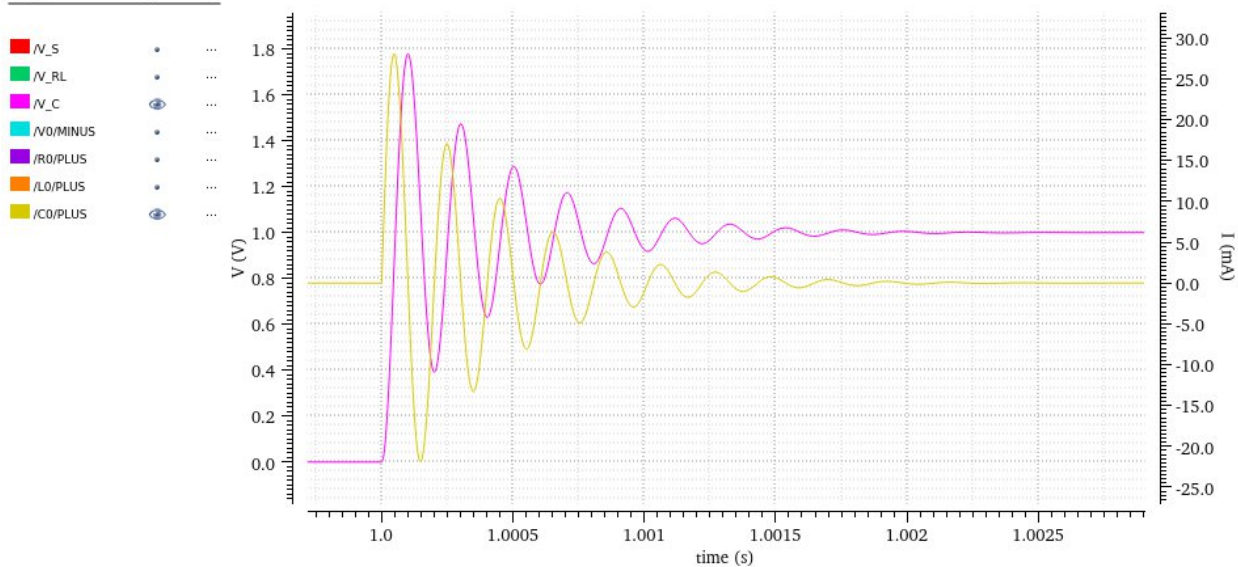


Figure 9. Graph for the Transient Response of the Capacitor Voltage and Current

With $R = 0\Omega$, the system exhibited undamped oscillations due to a lack of resistance.

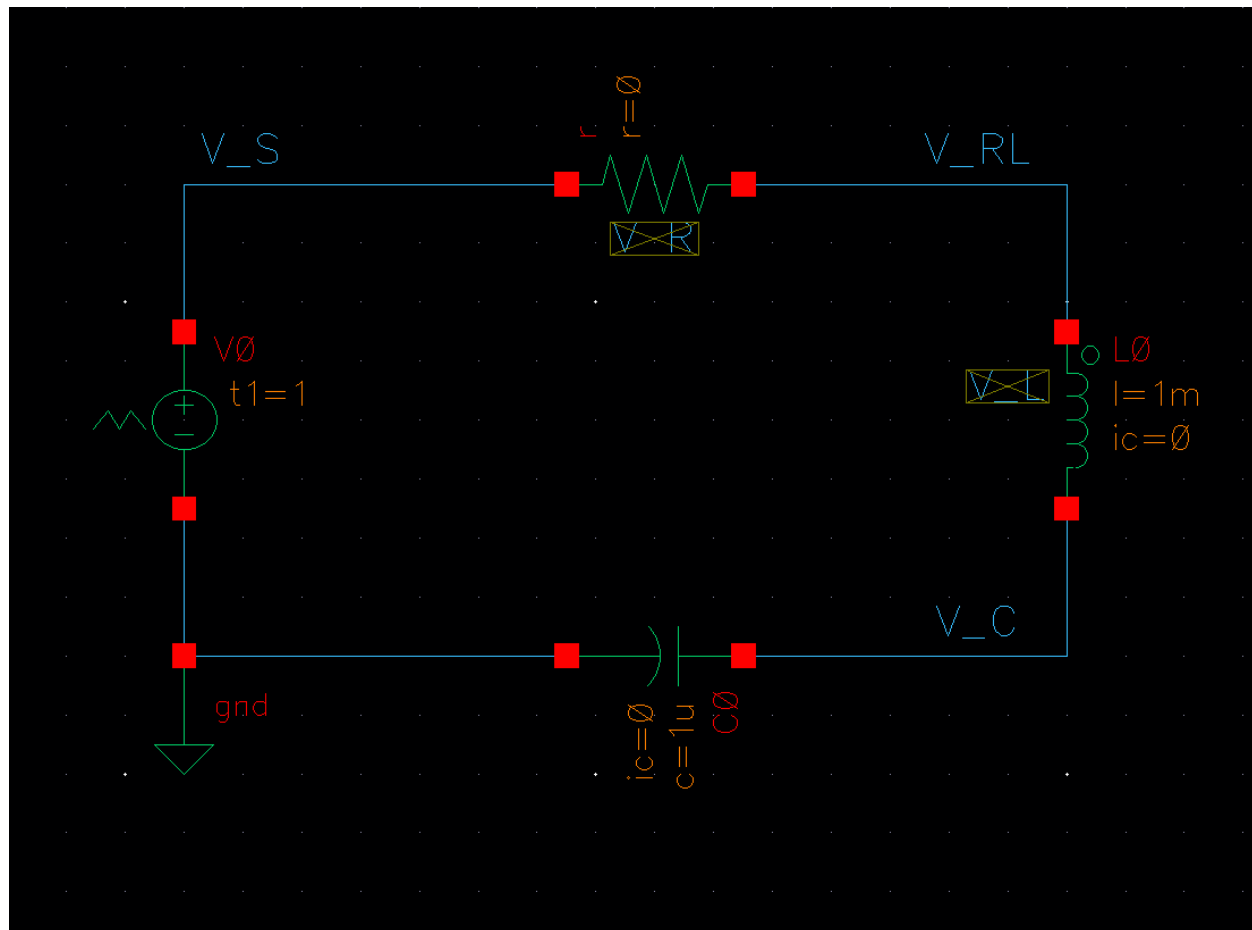


Figure 10: Schematic of series RLC circuit with 0 Ohm Resistor

Transient Response

Thu Feb 13 23:42:12
2025

1

Name ... t

- ☒ /V_S
- ☐ /V_RL
- ☐ /V_C
- ☐ /C0/PLUS
- ☐ /L0/PLUS
- ☐ /r/PLUS
- ☒ /V0/MINUS

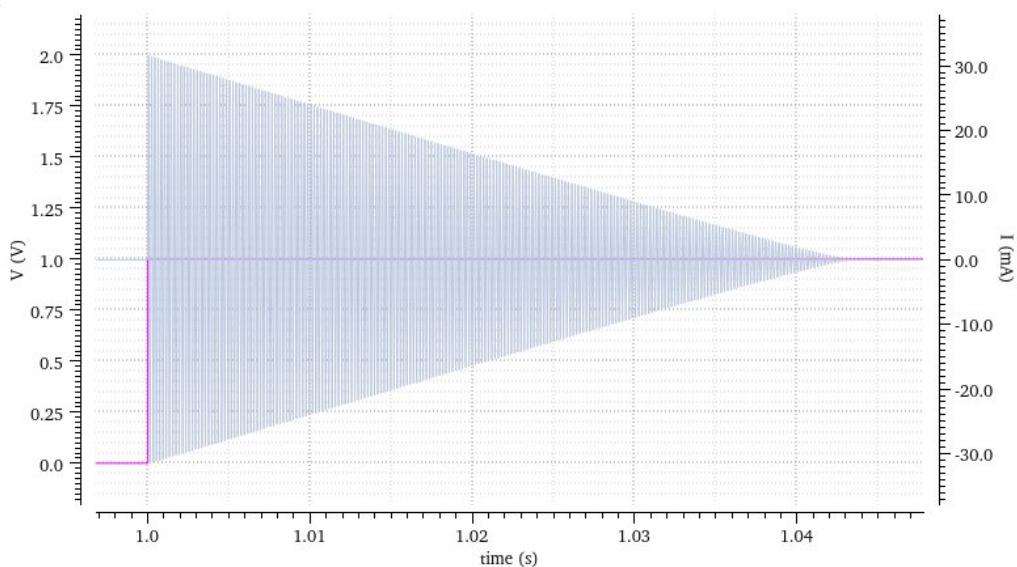


Figure 11: Graph for the Transient Response for the Source Voltage and Current

V_R

Thu Feb 13 23:50:07
2025

1

Name ... Design_Point

- ☒ V_R

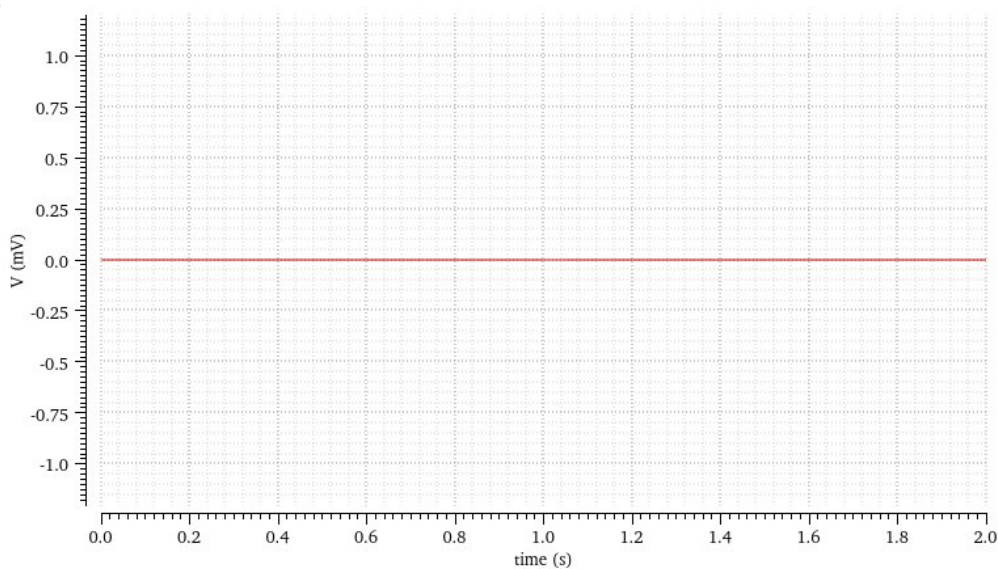


Figure 11: Graph for the Transient Response for the Resistor Voltage

Transient Response

Fri Feb 14 18:09:41
2025

1

Namet

- V_S • ...
- V_{RL} • ...
- V_C • ...
- $/C0/PLUS$ • ...
- $/L0/PLUS$ • ...
- $/r/PLUS$ ☒ • ...
- $/V0/MINUS$ • ...

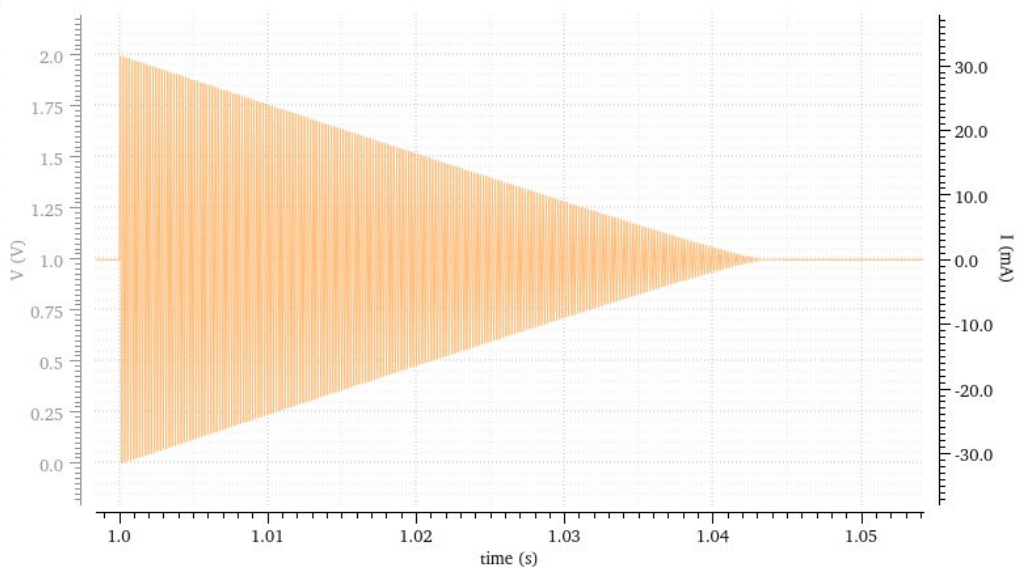


Figure 12: Graph for the Transient Response for the Resistor Current

Transient Response

Thu Feb 13 23:42:12
2025

1

Namet

- V_S ☒ • ...
- V_{RL} • ...
- V_C • ...
- $/C0/PLUS$ • ...
- $/L0/PLUS$ ☒ • ...
- $/r/PLUS$ • ...
- $/V0/MINUS$ • ...

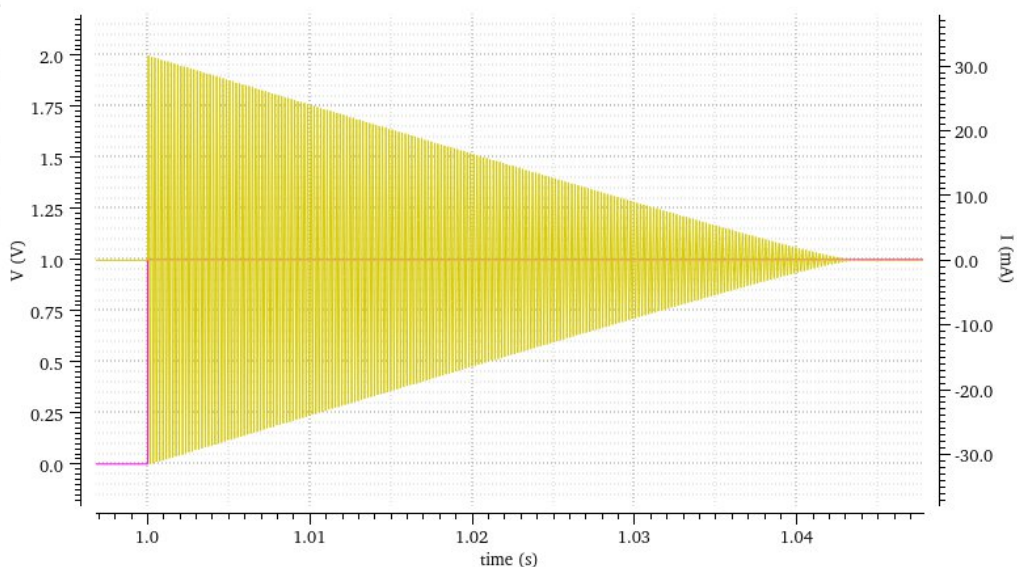


Figure 13: Graph for the Transient Response for the Inductor Current

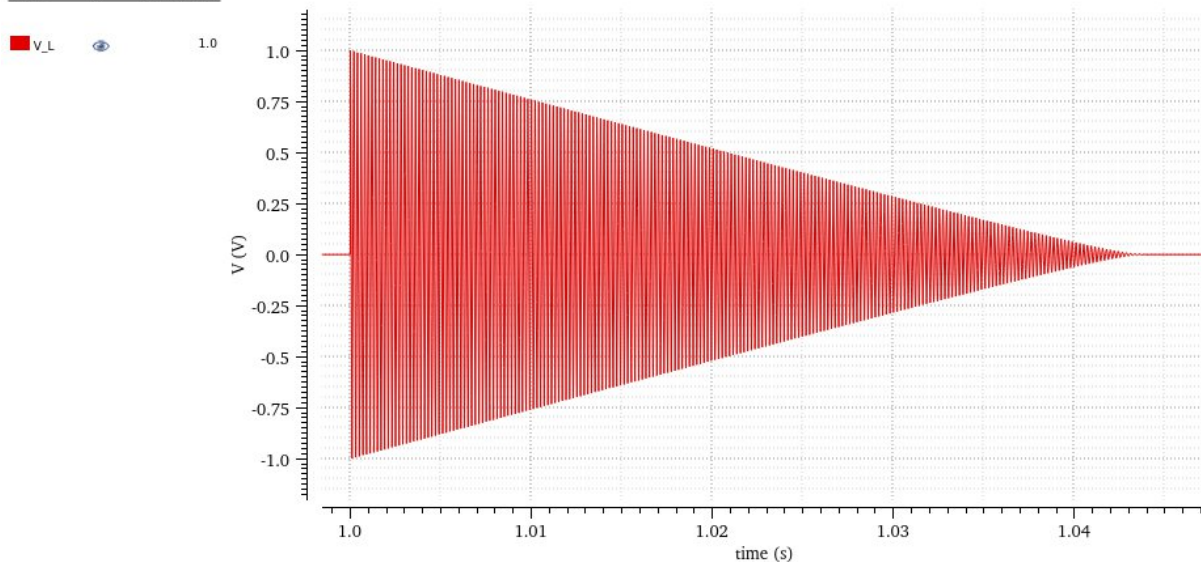


Figure 14: Graph for the Transient Response for the Inductor Voltage

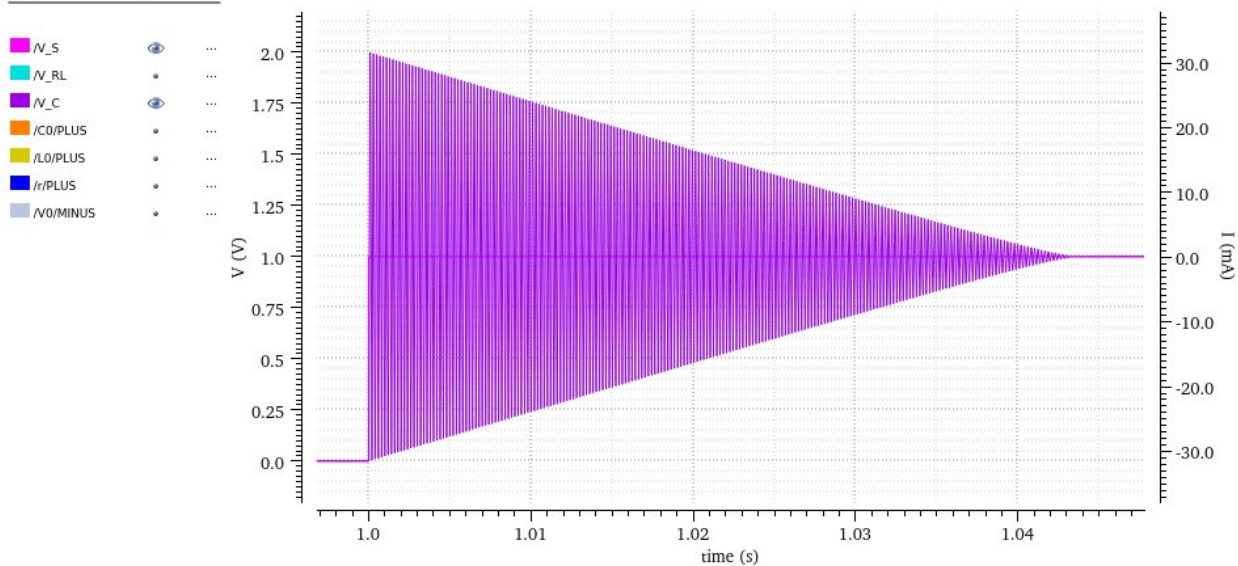


Figure 15: Graph for the Transient Response for the Voltage of the Capacitor

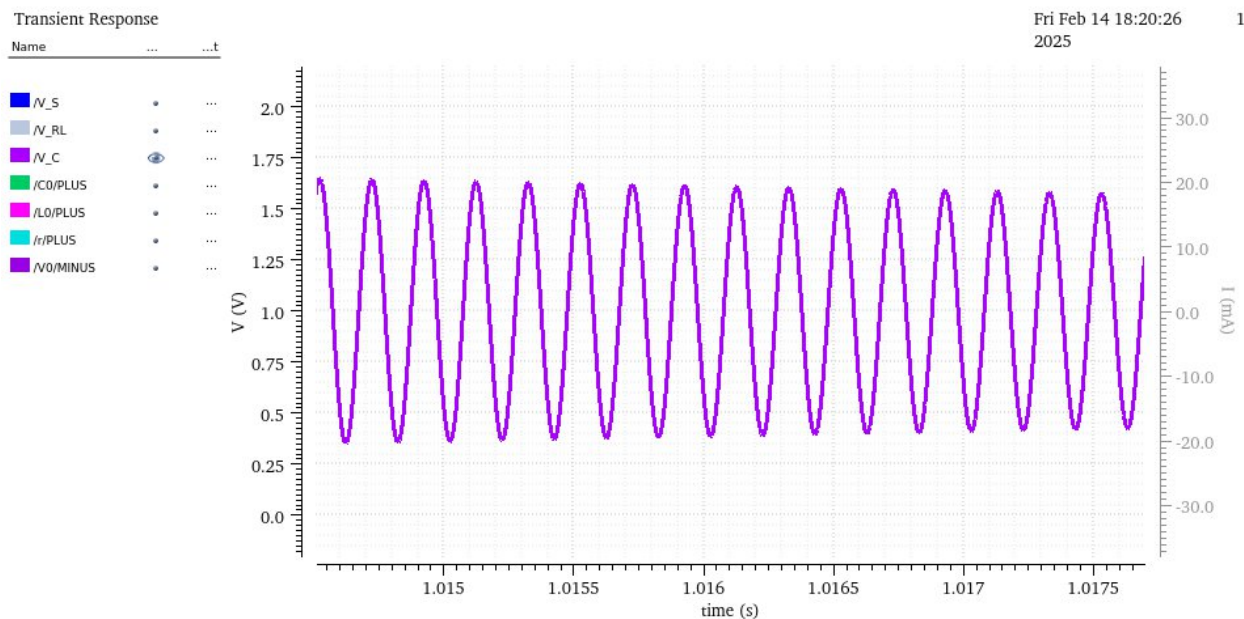


Figure 16: Graph for the Transient Response for Capacitor Voltage Zoomed

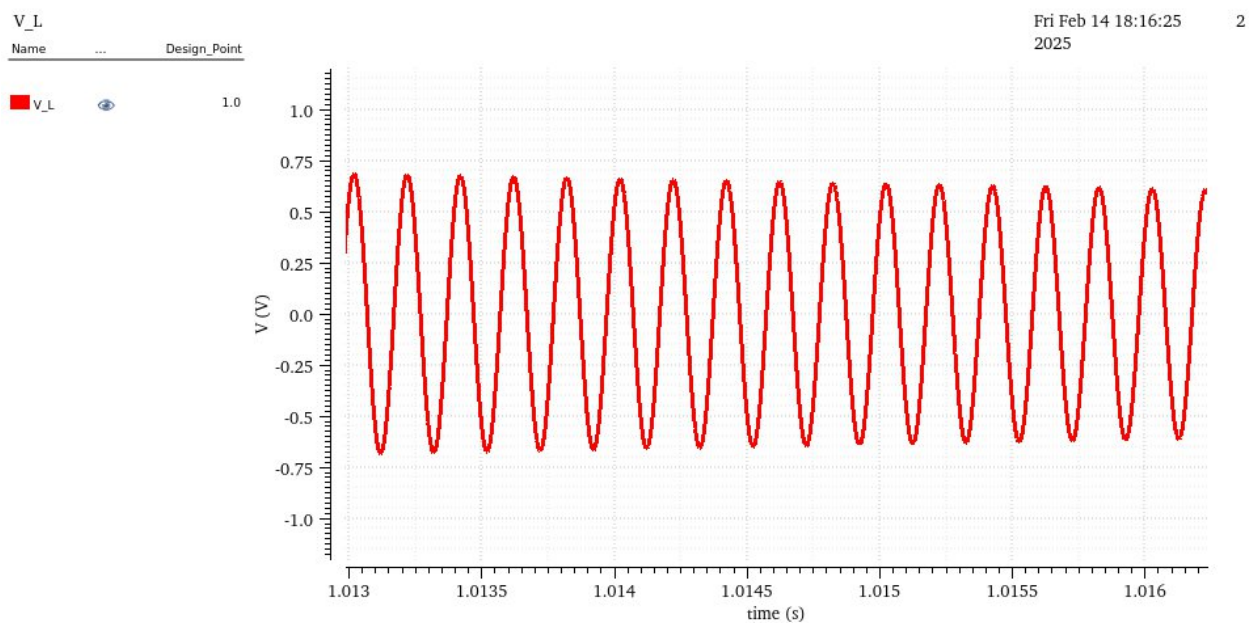


Figure 17: Graph for the transient response for the Voltage of the Inductor Zoomed

Circuit 2:

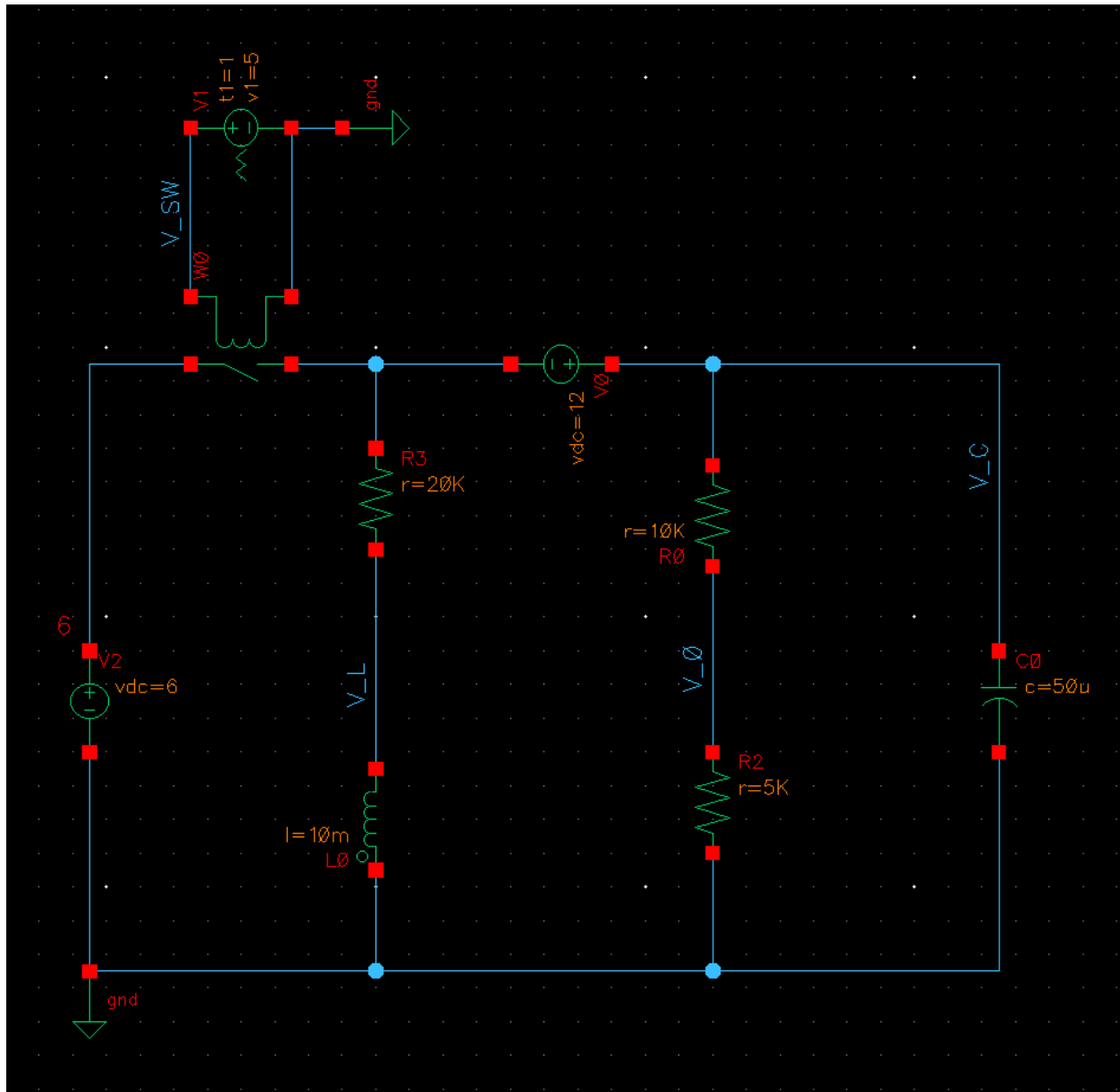


Figure 18: Cadence Schematic of RLC Circuit 2

Transient Response

Fri Feb 14 17:28:53
2025

1

Nameint

■ /V_L		1.0
■ /V_0		1.0
■ /V_C		1.0
■ /L0/PLUS		1.0
■ /R2/PLUS		1.0
■ /C0/PLUS		1.0

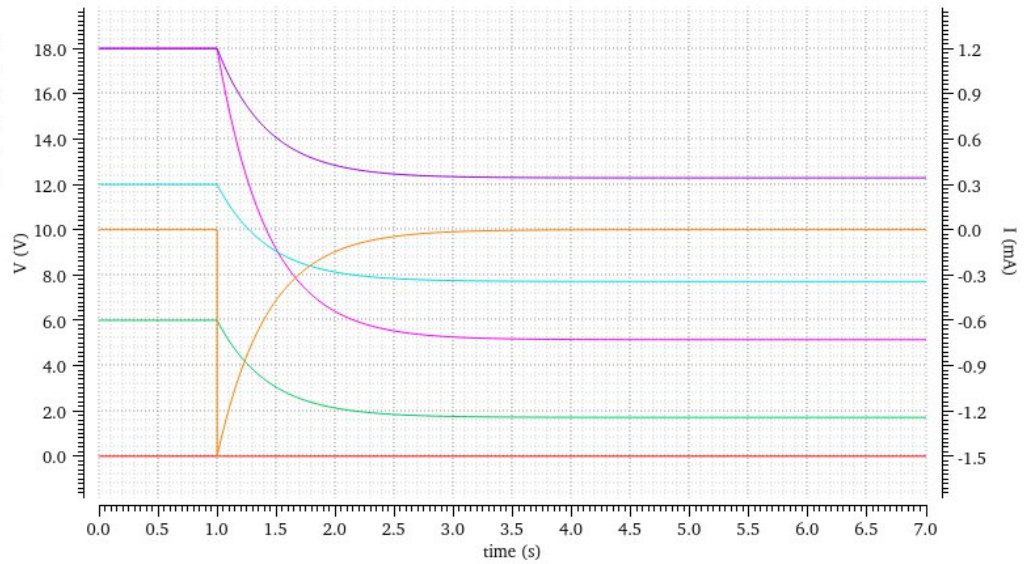


Figure 19: Graph showing the Transient Response for Resistor, Inductor and Capacitor

Transient Response

Fri Feb 14 17:28:53
2025

1

Nameint

■ /V_L		1.0
■ /V_0		1.0
■ /V_C		1.0
■ /L0/PLUS		1.0
■ /R2/PLUS		1.0
■ /C0/PLUS		1.0

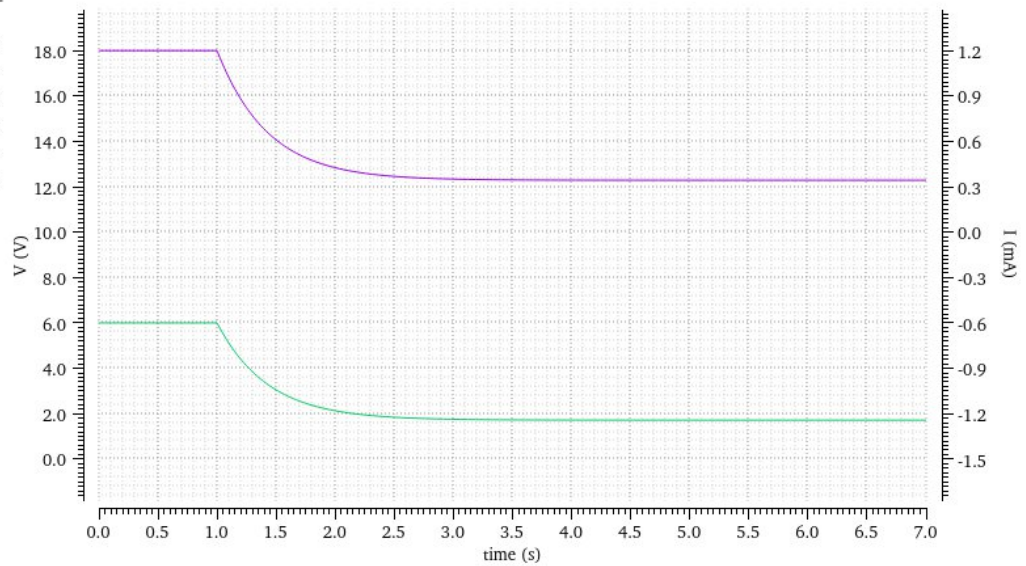


Figure 20: Graph for the Transient Response for the Voltage and the Current for the 5k Ohms Resistor

Transient Response

Fri Feb 14 17:28:53
2025

1

Nameint

■ /V_L		1.0
■ /V_0		1.0
■ /V_C		1.0
■ /I0/PLUS		1.0
■ /R2/PLUS		1.0
■ /C0/PLUS		1.0

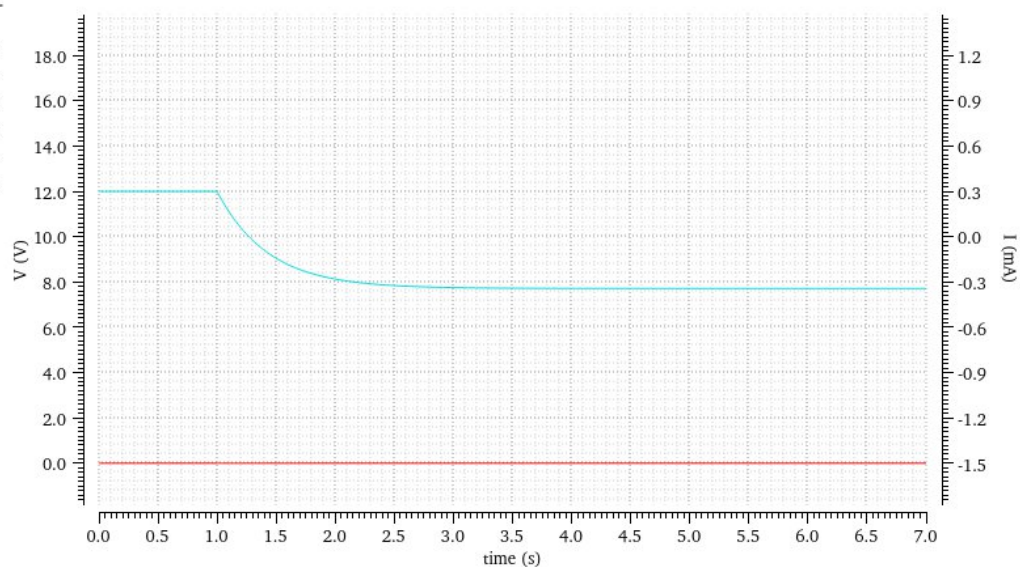


Figure 21: Graph for the Transient Response for the Voltage and the Current of the Inductor

Transient Response

Fri Feb 14 17:28:53
2025

1

Nameint

■ /V_L	•	1.0
■ /V_0	•	1.0
■ /V_C	•	1.0
■ /I0/PLUS	•	1.0
■ /R2/PLUS	•	1.0
■ /C0/PLUS	•	1.0

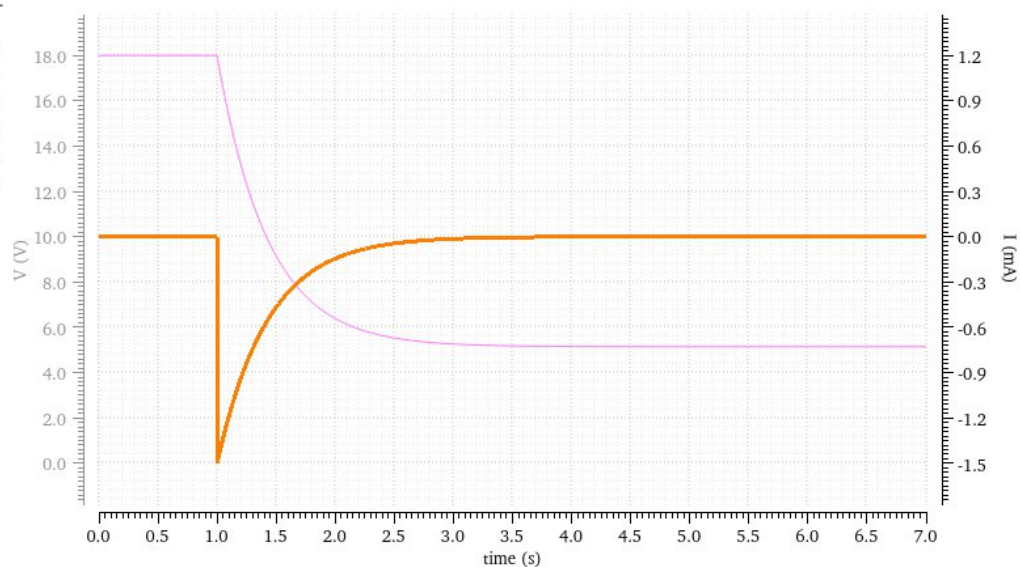


Figure 22: Graph for the Transient Response for the Voltage and Current of the Capacitor

Circuit 3:

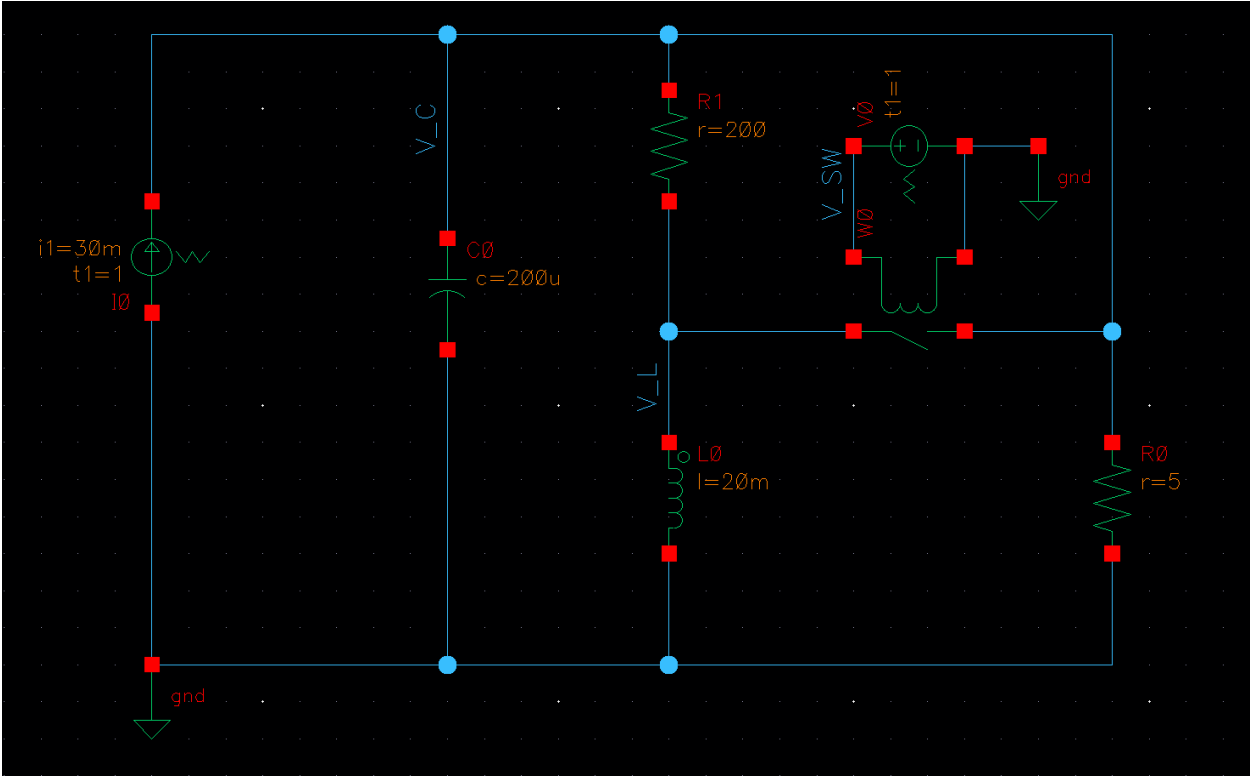


Figure 23: Cadence Schematic of RLC Circuit 3

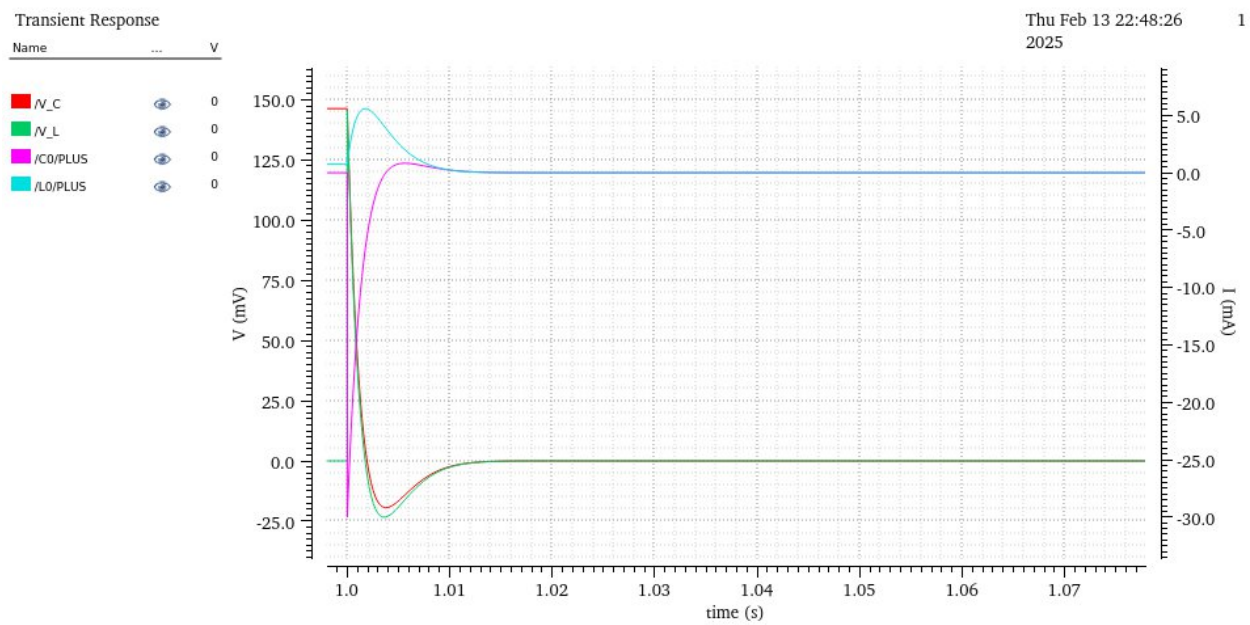


Figure 24: Graph for the Transient Response for Inductor and Capacitor

Transient Response

Thu Feb 13 22:48:26
2025

1

Name	---	V
------	-----	---

■ /V_C	•	0
■ /V_L	•	0
■ /C0/PLUS	•	0
■ /L0/PLUS	•	0

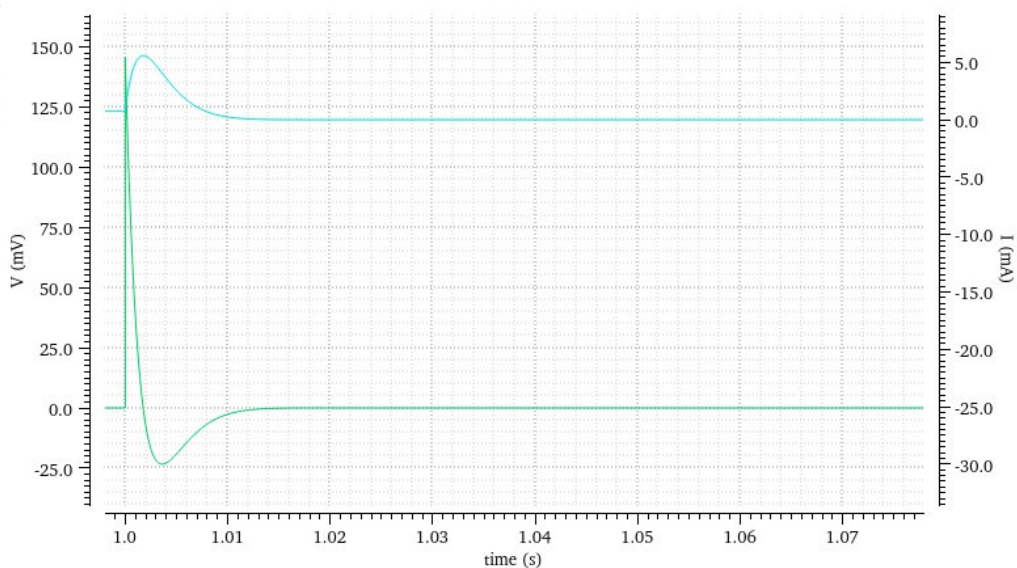


Figure 25: Graph for the Transient Response for the Voltage and Current of the Inductor

Transient Response

Thu Feb 13 22:48:26
2025

1

Name	---	V
------	-----	---

■ /V_C	•	0
■ /V_L	•	0
■ /C0/PLUS	•	0
■ /L0/PLUS	•	0

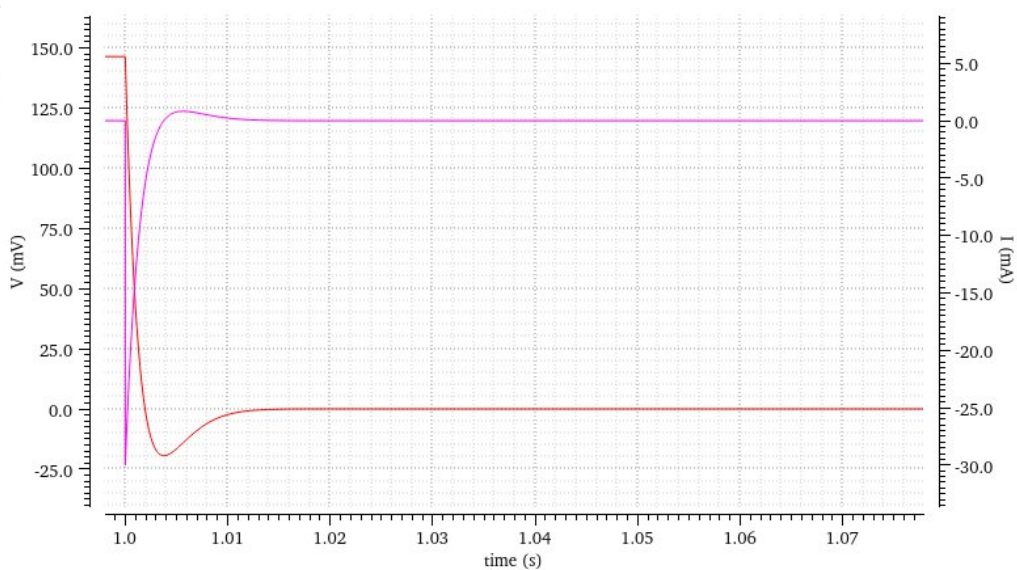


Figure 26: Graph for the Transient Response for Capacitor

Discussion/ Analysis of Results

Analysis of Transient Response from the Graphs Obtained from Circuit Schematic 1

Comparing the graphs for $R = 5 \text{ Ohms}$ and $R = 0 \text{ Ohms}$, the transient response settles much faster when a resistor is present. This is because the resistor dissipates energy as heat, damping the oscillations. With energy dissipation, the response stabilizes more quickly. When $R = 0 \text{ Ohms}$, the circuit oscillates indefinitely between the inductor and capacitor, resulting in a much longer transient period. Without a resistor, there is no mechanism to reduce the oscillations over time.

Analysis of Transient Response from the Graphs Obtained from Circuit Schematic 2

In the RLC circuit, the transient response depends on the damping factor. Since there are no oscillations, the circuit is likely overdamped or critically damped. Based on the plotted responses for the resistor, capacitor, and inductor:

1. Resistor ($5\text{k}\Omega$) Fig. 20

- Voltage (Green Curve): The voltage across the $5\text{k}\Omega$ resistor starts at $\sim 6\text{V}$, then **gradually decreases** and stabilizes at $\sim 1.5\text{V}$.
- Current (Purple Curve): The current initially peaks at around 1.2mA . It then **drops sharply** and continues decreasing exponentially until it stabilizes at a lower value ($\sim 0.3\text{mA}$).

2. Inductor (10mH) Fig. 21

- Voltage (Red Curve): The voltage across the inductor remained at 0V throughout the response.
- Current (Blue Curve): The inductor current starts at 0.3 A , then drops before stabilizing at -0.3 A . This suggests an initial energy source is pushing current through the inductor in one direction, but then the current reverses polarity before stabilizing due to the switch opening.

3. Capacitor ($50\mu\text{F}$) Fig. 22

- Voltage (Pink Curve): The capacitor voltage starts at approximately 18V and gradually decays as the capacitor discharges, eventually stabilizing at around 5V .
- Current (Orange Curve): The capacitor current initially spikes due to rapid discharge and then decreases until the capacitor is fully discharged.

Analysis of Transient Response from the Graphs Obtained from Circuit Schematic 3

1. Inductor Response Fig. 25

- At $t = 1\text{ s}$, the inductor voltage spikes to $\sim 150\text{ mV}$, followed by a negative dip before stabilizing. This occurs as the inductor resists sudden current changes by changing its polarity, leading to oscillatory energy exchange with the capacitor. The inductor current also spikes up to around 5 mA . Over time, the 5Ω resistor dissipates energy, damping the oscillations and ensuring a steady state where both the voltage and current stabilizes to zero.

2. Capacitor Response Fig. 26

- The capacitor voltage starts at $\sim 150\text{ mV}$, then drops sharply to a negative peak before settling. Initially, the capacitor holds charge but discharges through the circuit, causing oscillations due to interaction with the inductor. The capacitor current peaks at $\sim 125\text{ mV}$, then gradually stabilizes as resistive losses dampen the oscillations and bring the system to equilibrium.
- The current also shows this by spiking in the negative direction, reaching a peak of -30 mA before stabilising at 0.

It is observed from the graph consisting of both the inductor and capacitor plots in Figure 24 that both components exhibit the same settling time at around 1.014 seconds.

Conclusion

This lab improved our understanding of transient circuit behaviour using Cadence simulations. The observed responses aligned with theoretical predictions, reinforcing concepts of damping, oscillation, and settling time in RLC circuits.